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Grizon AI
Master Technical Document

Platform Audit · Legal Agent Architecture · Implementation Roadmap

PREPARED BY
Senior AI Systems Architect

DATE
August 7, 2026

PLATFORM VERSION
Grizon v0.4.0

SCOPE
Live Audit + Legal Agent + Roadmap

⚠️ Proprietary & Confidential — For Founder Use Only

> ℹ️ This document is a comprehensive technical audit prepared for the Grizon AI founding team. It combines a **live session analysis** of app.grizonai.com, review of all architecture documentation, and a deep analysis of the Legal Research & Case Analysis Agent. All findings are based on observed behavior — not assumptions.

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1. Executive Summary

Grizon AI is a **next-generation AI agent workspace platform** (v0.4.0) with a functioning core — authentication, streaming chat, PDF document analysis, credit billing via PhonePe UPI, and a well-designed multi-LLM routing engine called **Project Brain**.

However, the platform has **4 critical gaps** that must be resolved before serious growth:

#	Gap	Business Impact	Severity
1	**Bot doesn't know its own pricing** (RAG broken)	Every prospect who asks gets wrong answer — sales killer	CRITICAL
2	**Composer input disappears** after any response	Core UX broken — users can't have multi-turn conversations	CRITICAL

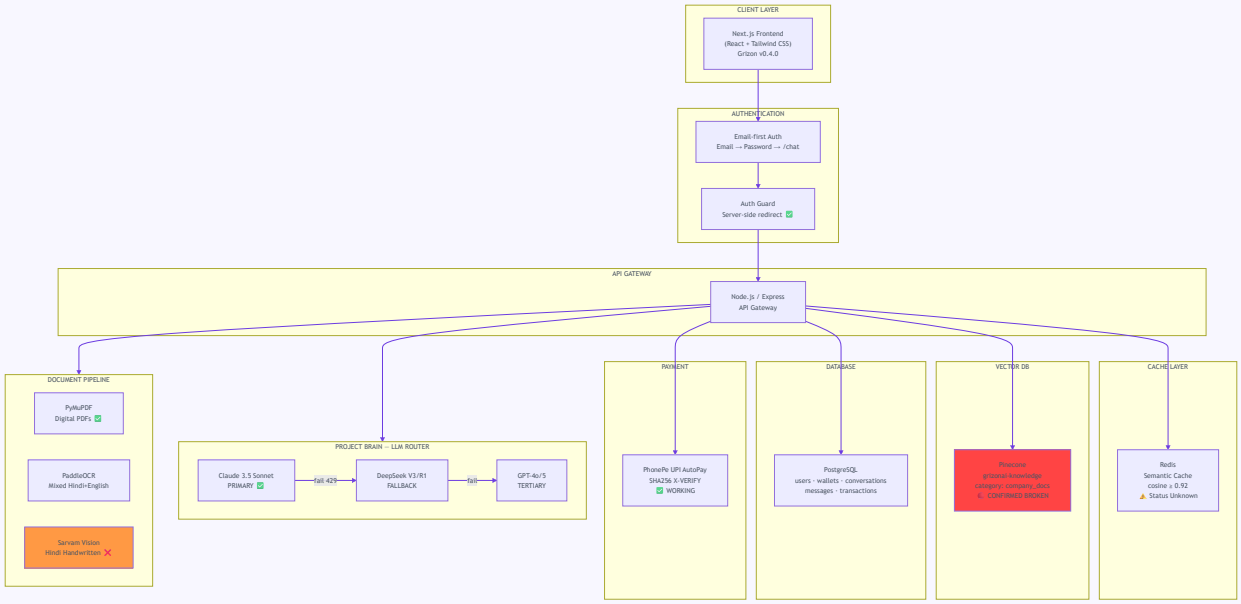
#	Gap	Business Impact	Severity
3	**Zero hallucination detection** active in production	Platform's core anti-hallucination promise not delivered	HIGH
4	**Legal Agent tools incomplete** (file_gen, Hindi OCR)	Legal product cannot launch without these	HIGH

📊 **Overall Platform Health: 5.5 / 10** | **Legal Agent Design Quality: 7 / 10** (excellent concept, tool gaps to resolve)

2. Current System Architecture

2.1 Live Confirmed Tech Stack

FIGURE 1 — LIVE CONFIRMED SYSTEM ARCHITECTURE



2.2 Working Features (Confirmed Live Session)

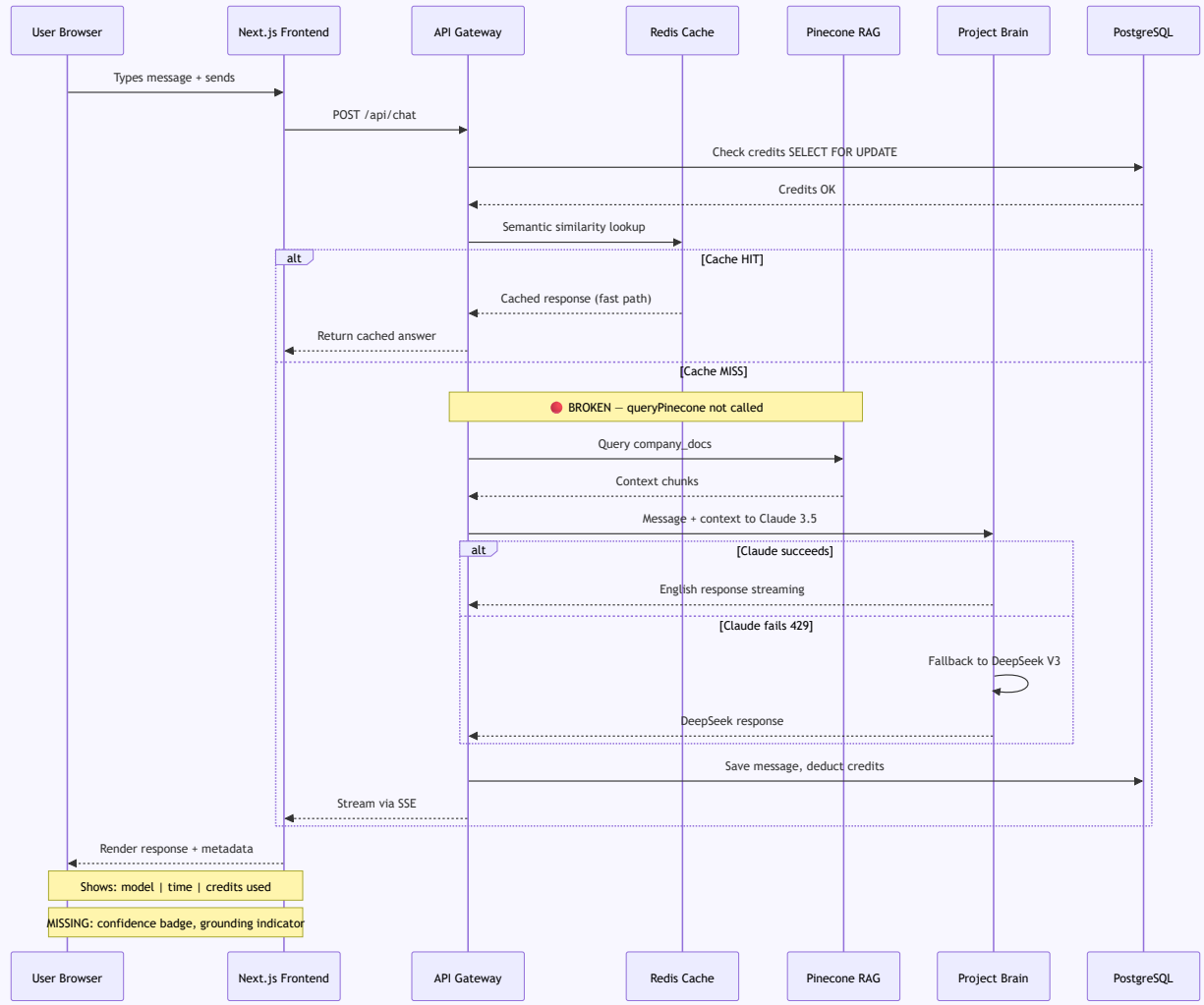
Component	Status	Performance Measured
User Authentication	✅ WORKING	Email → password, session persisted
Auth Guard	✅ WORKING	/chat → login redirect when logged out
Text Chat (streaming)	✅ WORKING	1.4s – 3.3s Time to First Token
PDF Document Upload + Analysis	✅ WORKING	5.7s, 661KB PDF, ZERO hallucination on doc content
Credit Deduction	✅ WORKING	3 credits/text, 6 credits/PDF query
Billing Dashboard	✅ WORKING	Pro ₹1,999/month, renews Aug 31 2026

Component	Status	Performance Measured
PhonePe UPI AutoPay	✅ WORKING	Mandate linked, recurring active
Model Selector	✅ WORKING	Auto mode + General Assistant dropdown
Rate Limiting	✅ WORKING	60/hr · 400/day · 2000/week · 10000/month
Mobile Responsive Layout	✅ WORKING	Hamburger drawer at mobile width
Token Streaming	✅ WORKING	Real-time token-by-token streaming

3. Data Flow Diagrams

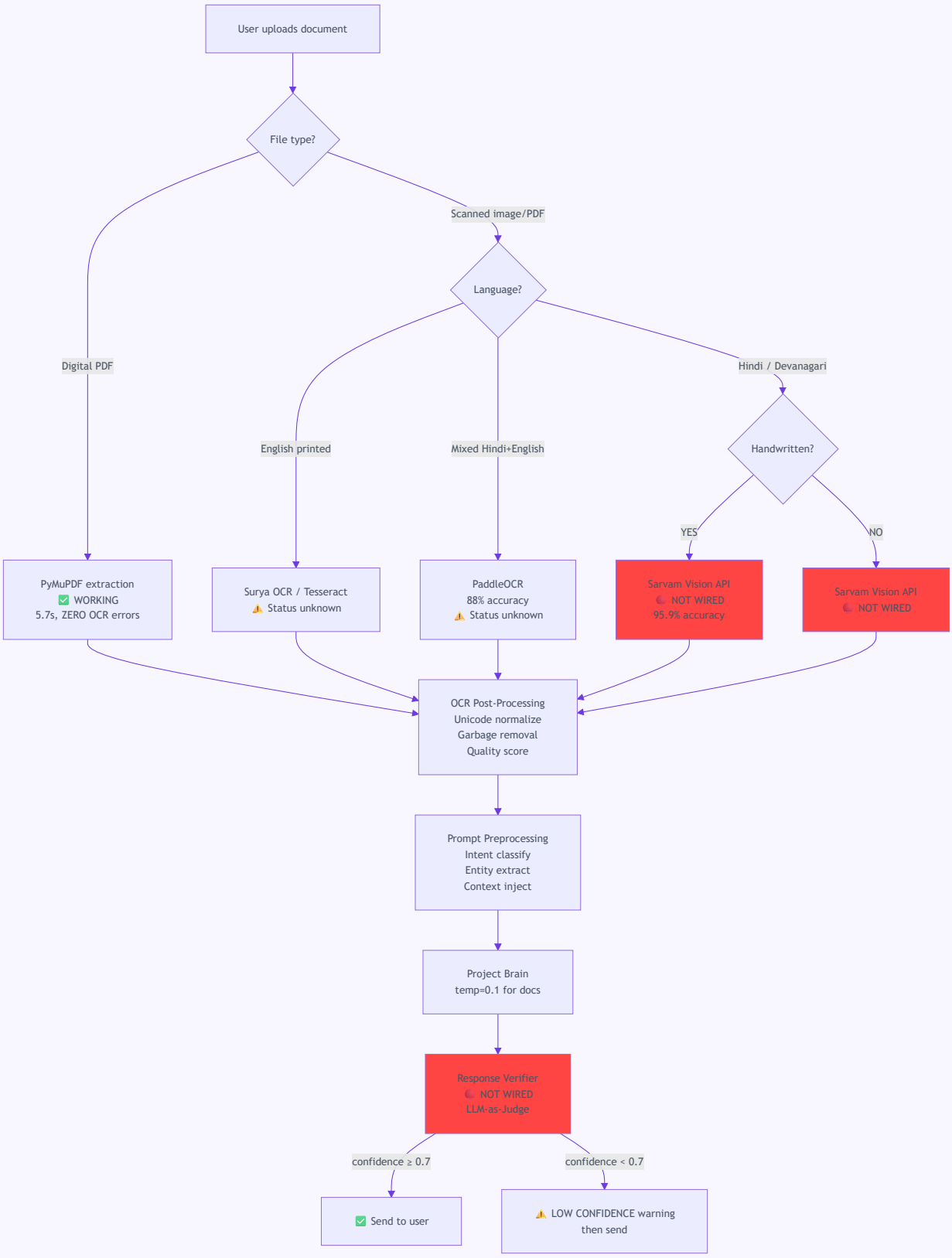
3.1 Complete Message Processing Flow

FIGURE 2 — MESSAGE PROCESSING SEQUENCE



3.2 Document / OCR Processing Pipeline

📐 FIGURE 3 — DOCUMENT PROCESSING DECISION TREE



4. System Health Scorecard

AUTH	CHAT	OCR/DOCS	HALLUCINATION	RAG/KNOWLEDGE
9	7	8	3	1
A	B	B+	D	F

MOBILE	PAYMENTS	VOICE	MULTILINGUAL	VERIFICATION
6	9	3	0	2
C+	A	D	N/A	F

📊 **Overall Score: 5.5 / 10 — Grade C+** — Strong payments & auth foundation; hallucination prevention and multilingual support are critical gaps holding the platform back.

Category	Score	Grade	Key Notes
Authentication	9/10	A	Solid email auth, server-side guard
Chat Functionality	7/10	B	Streaming works; composer push bug is critical
Document OCR (digital PDF)	8/10	B+	PyMuPDF excellent; Hindi OCR path missing
Hallucination Prevention	3/10	D	Architecture designed, NOT deployed in prod
Company Knowledge RAG	1/10	F	Pinecone broken — bot ignores own pricing
Mobile Experience	6/10	C+	Collapse works; iOS zoom risk; composer bug worse on mobile
Payment System	9/10	A	PhonePe AutoPay fully working end-to-end
Voice Input	3/10	D	UI stub only — no Sarvam STT wired
Multilingual Support	0/10	N/A	English only; Sarvam not integrated

Category	Score	Grade	Key Notes
Response Verification	2/10	F	ResponseVerifier built but not active
OVERALL	**5.5/10**	**C+**	Strong foundation, critical gaps remain

5. Critical Bugs — Platform Core

CRITICAL — SALES KILLER

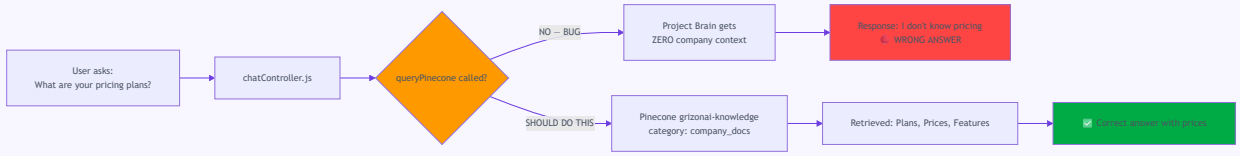
BUG #1 — Bot Doesn't Know Its Own Pricing

What happened live: Asked "Tell me about your pricing plans" → Bot responded: *"I don't currently have public pricing plans to share."* — while the user IS on the Pro ₹1,999/month plan.

Root Cause: `chatController.js` does not call `queryPinecone()` for general chat queries. The Pinecone `grizonai-knowledge` index is either empty or bypassed.

Fix: Populate Pinecone with company docs (pricing, features, FAQ) and wire `queryPinecone({ category: 'company_docs' })` to every chat request.

FIGURE 4 — BUG #1 ROOT CAUSE FLOW



CRITICAL — CORE UX BROKEN

BUG #2 — Composer Input Disappears After Any Long Response

What happened: After the first long response (5+ bullet points), the chat message area grows and pushes the composer input box off-screen (Y=945px → Y=1400+). Users cannot type follow-up messages.

Root Cause: Page-level scroll is active instead of the messages area having its own `overflow-y: auto`. The flex layout is not pinning the composer to the bottom.

Fix: Apply `flex: 1; overflow-y: auto; min-height: 0` to messages container. Apply `flex-shrink: 0` to composer. Use `height: 100dvh` on page wrapper.

MEDIUM

BUG #3 — Voice Textarea Hidden After Recording Stops

What happened: After stopping voice recording, the textarea stays hidden. User must click blank space to recover it.

Root Cause: Voice mode hides textarea, shows recording UI. On stop, state reverts but focus is not restored.

Fix (1 line): `setTimeout(() ⇒ textareaRef.current?.focus(), 150)` after recording stops.

HIGH — PROMISE NOT DELIVERED

BUG #4 — Response Verifier / LLM-as-Judge NOT Active

From architecture docs: 8-stage pipeline includes Stage 7: Response Verification with LLM-as-Judge, confidence scoring, number-checking.

From live observation: Response footer shows "General Assistant | 1.4s | 3 credits used" — NO confidence rating, NO [HIGH/MEDIUM/LOW CONFIDENCE] badge, NO grounding indicator. Bot confidently gave WRONG pricing info with zero flag.

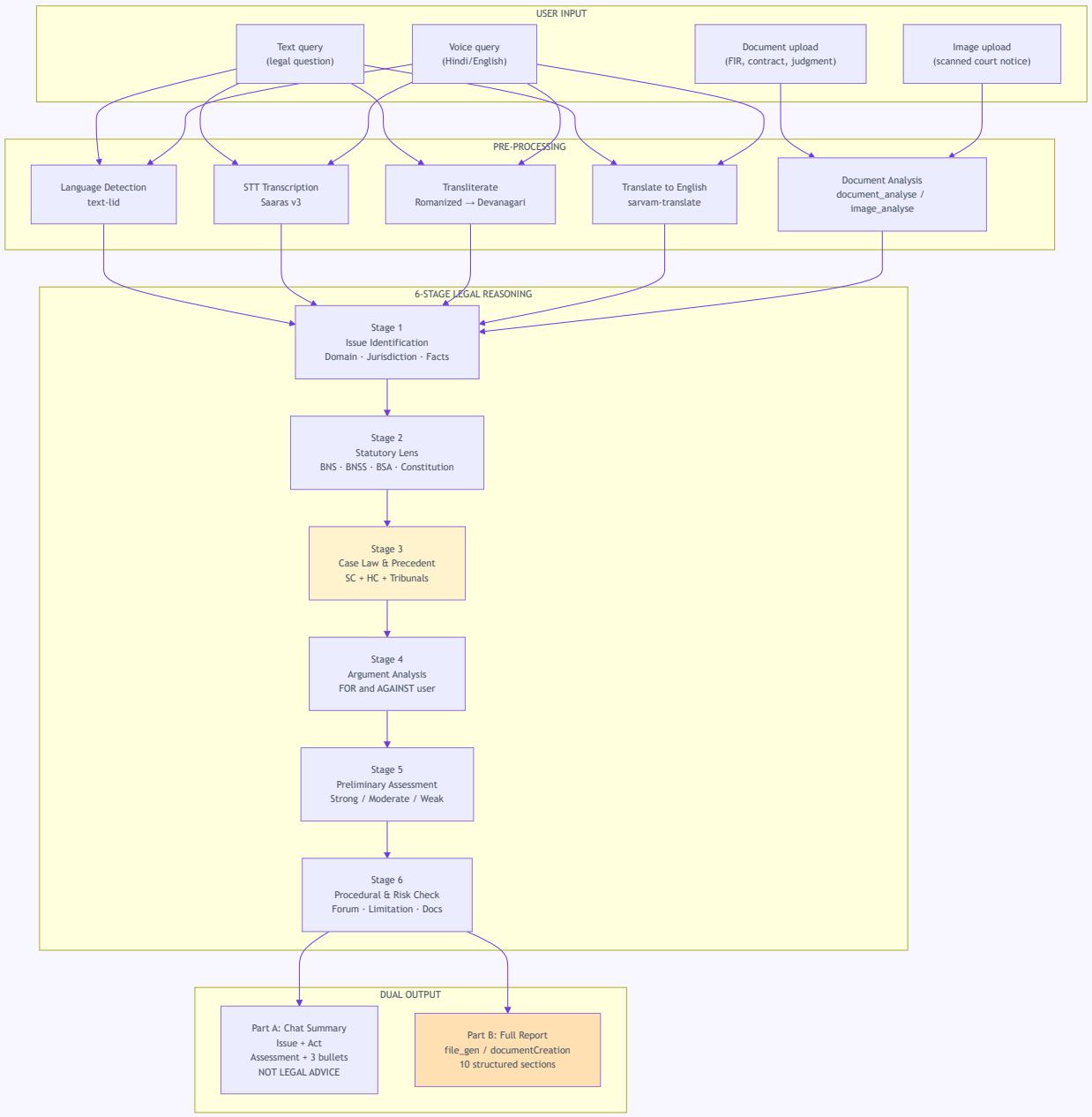
Fix: Wire `ResponseVerifier` class to chatController. Return `{ confidence, verified, grounded }` to frontend. Display confidence badge on every message.

6. Legal Research Agent Architecture

6.1 What This Agent Is

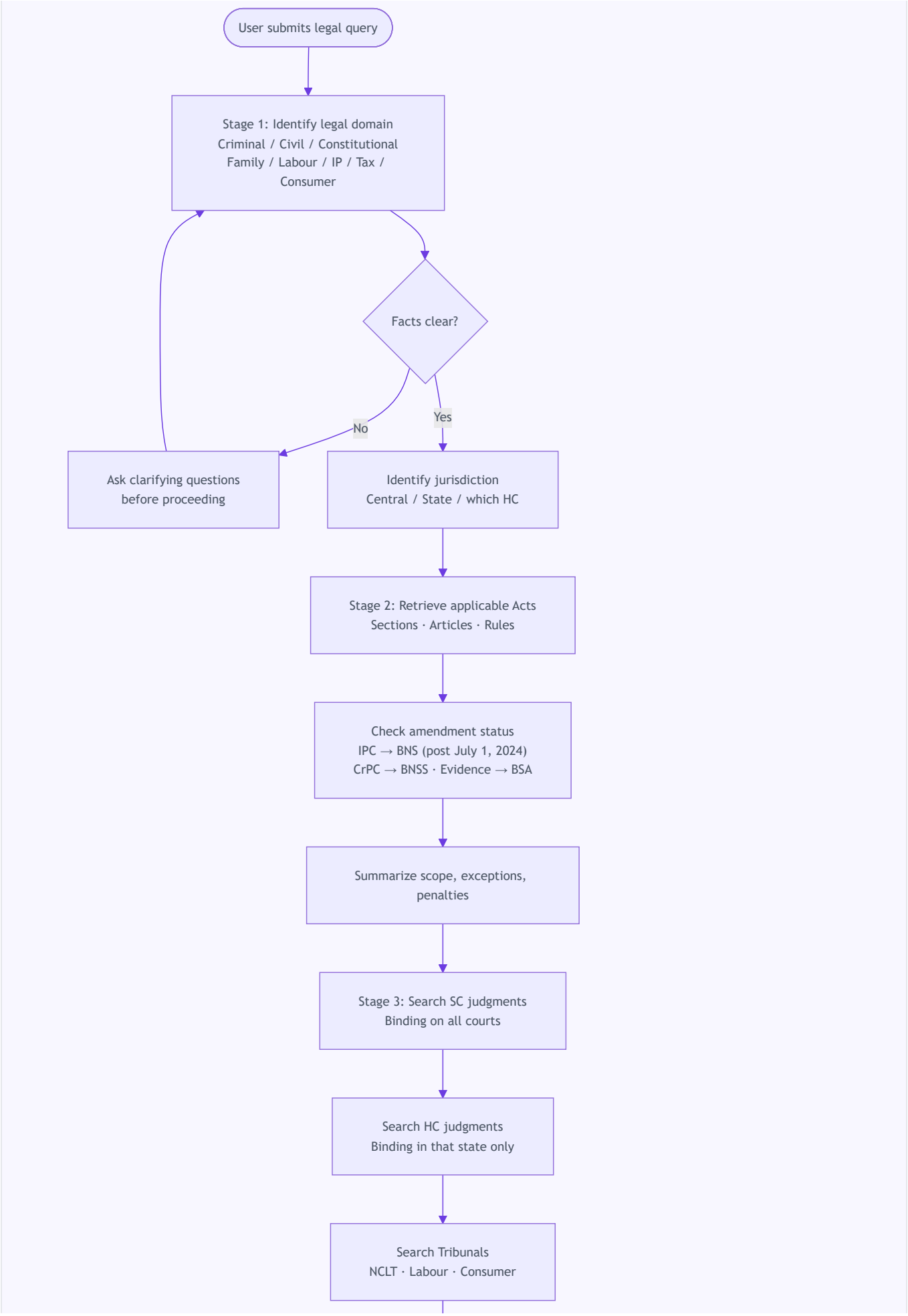
The **Legal Research & Case Analysis Agent** is a specialized AI persona layered on top of Project Brain. It transforms Grizon AI into a structured 6-stage legal reasoning engine purpose-built for **Indian law** (BNS, BNSS, BSA, Constitution of India, High Court + Supreme Court precedents).

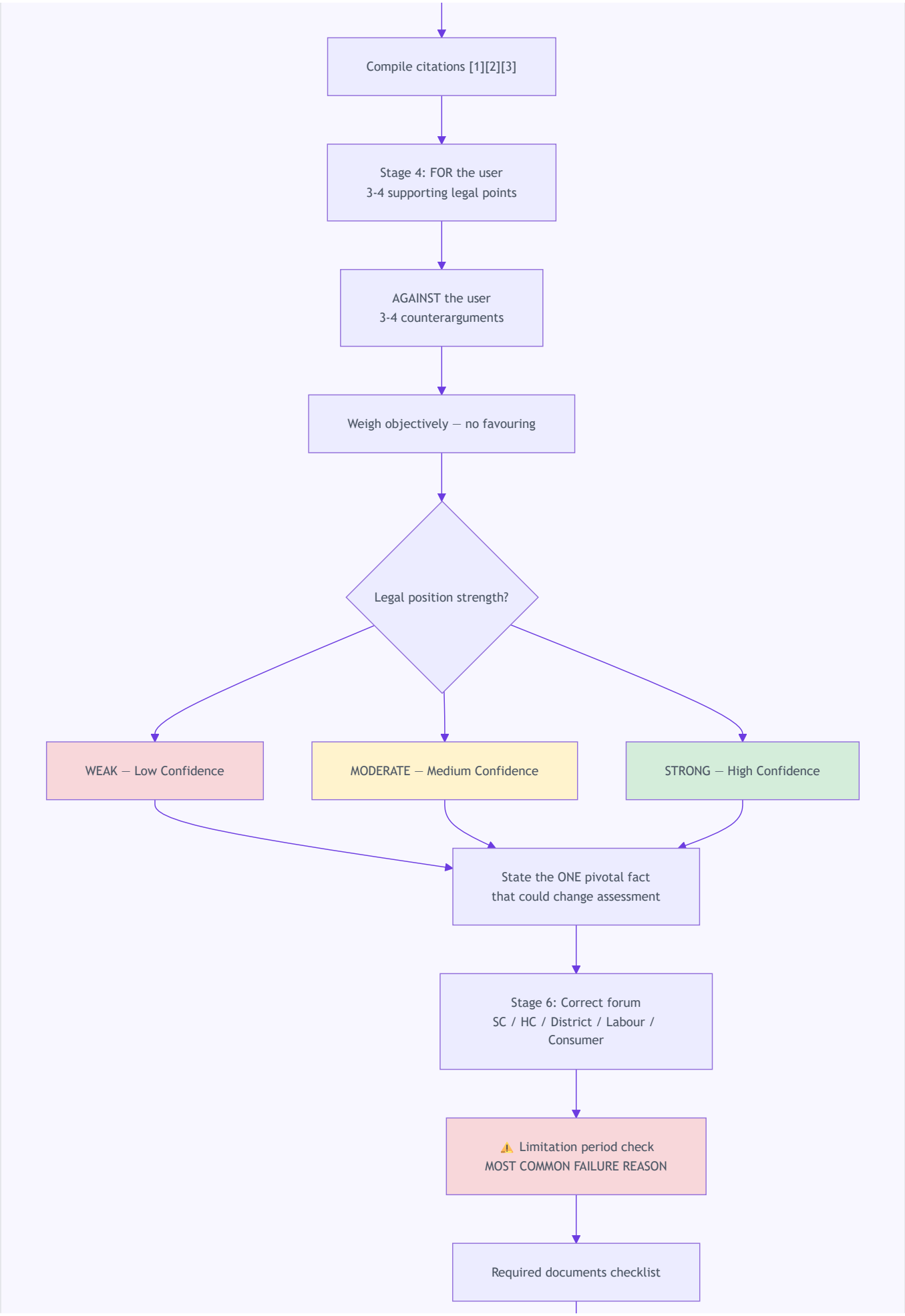
FIGURE 5 — LEGAL AGENT COMPLETE ARCHITECTURE

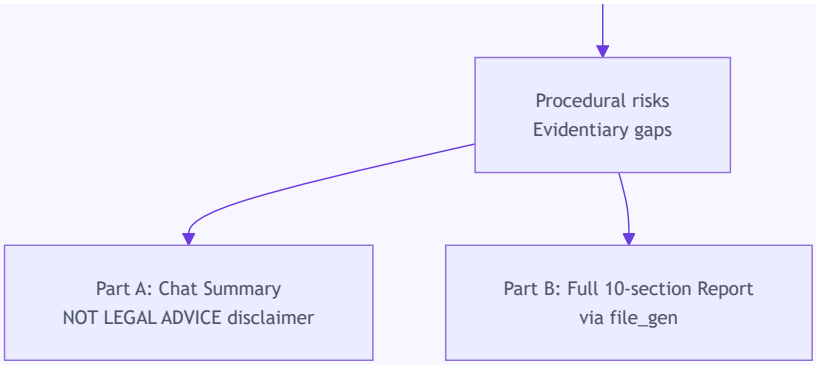


6.2 The 6-Stage Workflow — Detailed Flowchart

📐 FIGURE 6 — 6-STAGE LEGAL REASONING WORKFLOW


User submits legal query
Stage 1: Identify legal domain
Criminal / Civil / Constitutional
Family / Labour / IP / Tax /
Consumer
Facts clear?
No
Yes
Ask clarifying questions
before proceeding
Identify jurisdiction
Central / State / which HC
Stage 2: Retrieve applicable Acts
Sections · Articles · Rules
Check amendment status
IPC → BNS (post July 1, 2024)
CrPC → BNSS · Evidence → BSA
Summarize scope, exceptions,
penalties
Stage 3: Search SC judgments
Binding on all courts
Search HC judgments
Binding in that state only
Search Tribunals
NCLT · Labour · Consumer


Compile citations [1][2][3]
Stage 4: FOR the user
3-4 supporting legal points
AGAINST the user
3-4 counterarguments
Weigh objectively — no favouring
Legal position strength?
WEAK — Low Confidence
MODERATE — Medium Confidence
STRONG — High Confidence
State the ONE pivotal fact
that could change assessment
Stage 6: Correct forum
SC / HC / District / Labour /
Consumer
⚠️ Limitation period check
MOST COMMON FAILURE REASON
Required documents checklist



6.3 Tool Requirements vs. Current Status

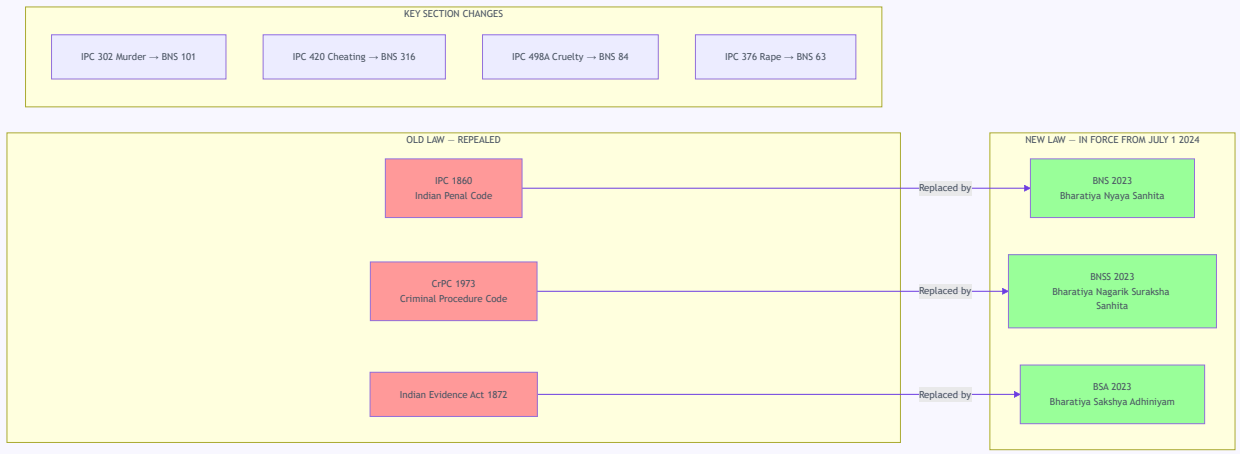
Tool	Purpose	Status	Risk if Missing
`document_analyse`	Read FIRs, contracts, judgments, PDFs	Partially Working	Medium — scanned docs fail
`web_search`	Retrieve statutes, case law, notifications	Present	High — must reach verified sources
`image_analyse`	Read scanned court notices, handwritten docs	❌ Not wired	HIGH — most Indian legal docs are scanned
`file_gen / documentCreation`	Generate Part B full report document	❌ Not confirmed	CRITICAL — entire Part B output fails

7. Legal Agent Risk Analysis

⚖️ **Legal AI Hallucination is the most dangerous category of AI error.** A fabricated legal citation in an Indian court filing is worse than no citation at all. The agent's "never fabricate" rule only works if verified legal databases (Indian Kanoon, SCC Online) are actually connected.

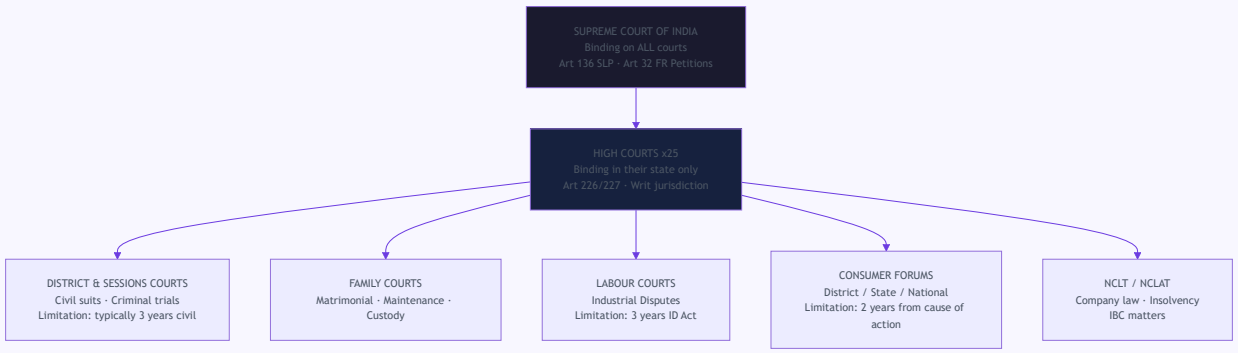
7.1 IPC → BNS Transition — Critical Context

📐 FIGURE 7 — INDIA'S CRIMINAL LAW TRANSITION (EFFECTIVE JULY 1, 2024)



7.2 Indian Court Hierarchy & Limitation Periods

FIGURE 8 — INDIAN COURT HIERARCHY



7.3 Top 5 Legal Agent Risks

#	Risk	Severity	Mitigation
1	**Legal Citation Hallucination** — LLM fabricates plausible case names	CRITICAL	Connect Indian Kanoon API; mandate source citation only
2	**IPC→BNS Confusion** — Wrong section numbers for post-July 2024 cases	CRITICAL	Date-of-offence detection; auto-switch law framework
3	**file_gen Failure** — Part B document never generated	HIGH	Implement file_gen as proper tool call
4	**Disclaimer Missing** — "Not legal advice" missed in any response	HIGH	Hard-code as system prompt footer — not soft instruction
5	**Hindi Doc OCR Failure** — FIRs, notices in Hindi unreadable	MEDIUM	Wire Sarvam Vision API for scanned Hindi documents

8. Feature Implementation Status

Feature	Category	Status	Notes
Email Authentication	Core	✅ Working	Confirmed live
Auth Guard (server-side)	Core	✅ Working	Confirmed live
Text Chat + Streaming	Core	✅ Working	1.4-3.3s TTFT
Digital PDF Extraction	Documents	✅ Working	PyMuPDF, excellent accuracy
Credit Deduction System	Billing	✅ Working	3/6 credits per query type
PhonePe UPI AutoPay	Billing	✅ Working	Mandate linked, recurring active
Billing Dashboard	Billing	✅ Working	Plan, credits, history visible
Model Selector (Auto)	AI	✅ Working	Claude/DeepSeek routing active
Rate Limiting	Core	✅ Working	60/hr, 400/day, 2k/wk, 10k/mo
Mobile Responsive Layout	UI	✅ Working	Hamburger at mobile width
Pinecone RAG — Company Docs	AI	🔴 Broken	Bot ignores own pricing
Composer Push Bug	UI	🔴 Broken	Input disappears after response
Response Verifier / LLM-Judge	AI	🔴 Broken	Built but not wired in production
Voice Input (Sarvam STT)	Voice	🟡 Stub	UI exists, no Sarvam Saaras v3
Scanned Hindi OCR	Documents	🟡 Partial	Sarvam Vision not connected
Redis Semantic Cache	Performance	🟡 Unknown	Server-side, status unconfirmed
Confidence Badges UI	UI	🟡 Partial	API returns data, frontend ignores it
Sarvam TTS (Bulbul v2)	Voice	❌ Not Built	Not started
Hindi / Multilingual Support	Languages	❌ Not Built	English only currently

Feature	Category	Status	Notes
file_gen Document Creation	Legal Agent	❌ Not Built	Required for Legal Agent Part B
Indian Kanoon API	Legal Agent	❌ Not Built	Critical for case law retrieval
Legal RAG (BNS/BNSS/BSA/Const)	Legal Agent	❌ Not Built	Foundation for Legal Agent
IPC→BNS Date Detection	Legal Agent	❌ Not Built	Critical legal accuracy
PhonePe Voice Checkout	Voice	❌ Not Built	Designed in architecture docs
WhatsApp Integration	Channels	❌ Not Built	Planned, not started

9. Priority Implementation Roadmap

FIGURE 9 — PRIORITY MATRIX (EFFORT VS. BUSINESS IMPACT)



Week-by-Week Execution Plan

WEEK 1 — STOP THE BLEEDING

Fix Composer Push Bug

CSS flex fix · pin composer to bottom · 2 hours · Zero risk · Every conversation becomes usable

🧠 **Populate Pinecone + Wire RAG**

Add pricing, features, FAQ docs to grizonai-knowledge · wire queryPinecone() to chatController · 1 day

🎤 **Fix Voice Textarea Focus Bug**

Add setTimeout focus restore after recording stops · 1 hour

📱 **Fix iOS Font-Size Zoom**

Set textarea font-size: 16px on mobile breakpoint · 30 minutes

WEEK 2 — HALLUCINATION PREVENTION

⚖️ **Wire ResponseVerifier to Chat Flow**

LLM-as-Judge + number-check + groundedness score · return confidence to frontend

🏷️ **Add Confidence Badges to Frontend**

Show [HIGH/MEDIUM/LOW CONFIDENCE] on every response · display grounding indicator for docs

🌡️ **Task-Adaptive Temperature**

temp=0.1 for document Q&A · temp=0.7 for general chat · detect based on document context

💳 **Fix Two Credit Pool UX Confusion**

Add tooltip explaining "Plan credits" vs "Wallet credits remaining"

WEEK 3 — LEGAL AGENT FOUNDATION

⚖️ **Integrate Indian Kanoon API**

Wire to web_search for verified SC + HC case law retrieval · prevents citation hallucination

📚 **Build Legal RAG in Pinecone**

Ingest: BNS 2023 · BNSS 2023 · BSA 2023 · Constitution of India · landmark judgments · category: legal_docs

📄 **Implement file_gen Tool**

Wire document generation as real tool call · output structured PDF/markdown for Legal Agent Part B

📅 **IPC→BNS Date Detection**

Auto-detect offence date · apply IPC if pre-July 1 2024 · apply BNS if after

WEEK 4 — SARVAM AI INTEGRATION (OPENS 1.4B USERS)

🎤 **Wire Saaras v3 STT to Voice Button**

Replace stub with real Sarvam STT API · Hinglish codemix mode · Hindi legal queries via voice

🔊 **Wire Bulbul v2 TTS Audio Output**

Add audio playback option · abhilash voice for AI assistant · base64 WAV output

🌐 **Language Detection + Translation Pipeline**

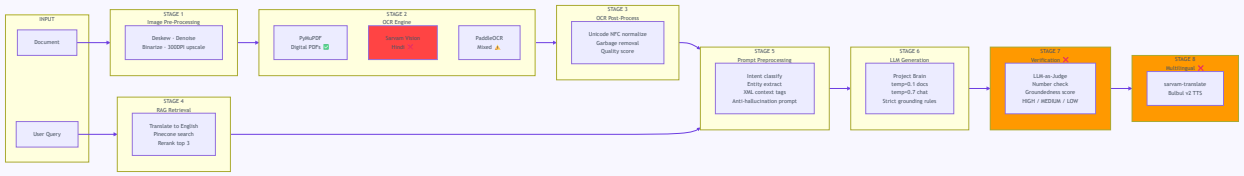
text-lid on every incoming message · translate non-English to English for RAG/LLM · translate response back

📷 **Wire Sarvam Vision OCR for Hindi Docs**

Enable Hindi FIR / court notice / handwritten document analysis for Legal Agent

10. Anti-Hallucination Architecture

📐 FIGURE 10 — COMPLETE 8-STAGE ZERO-HALLUCINATION PIPELINE

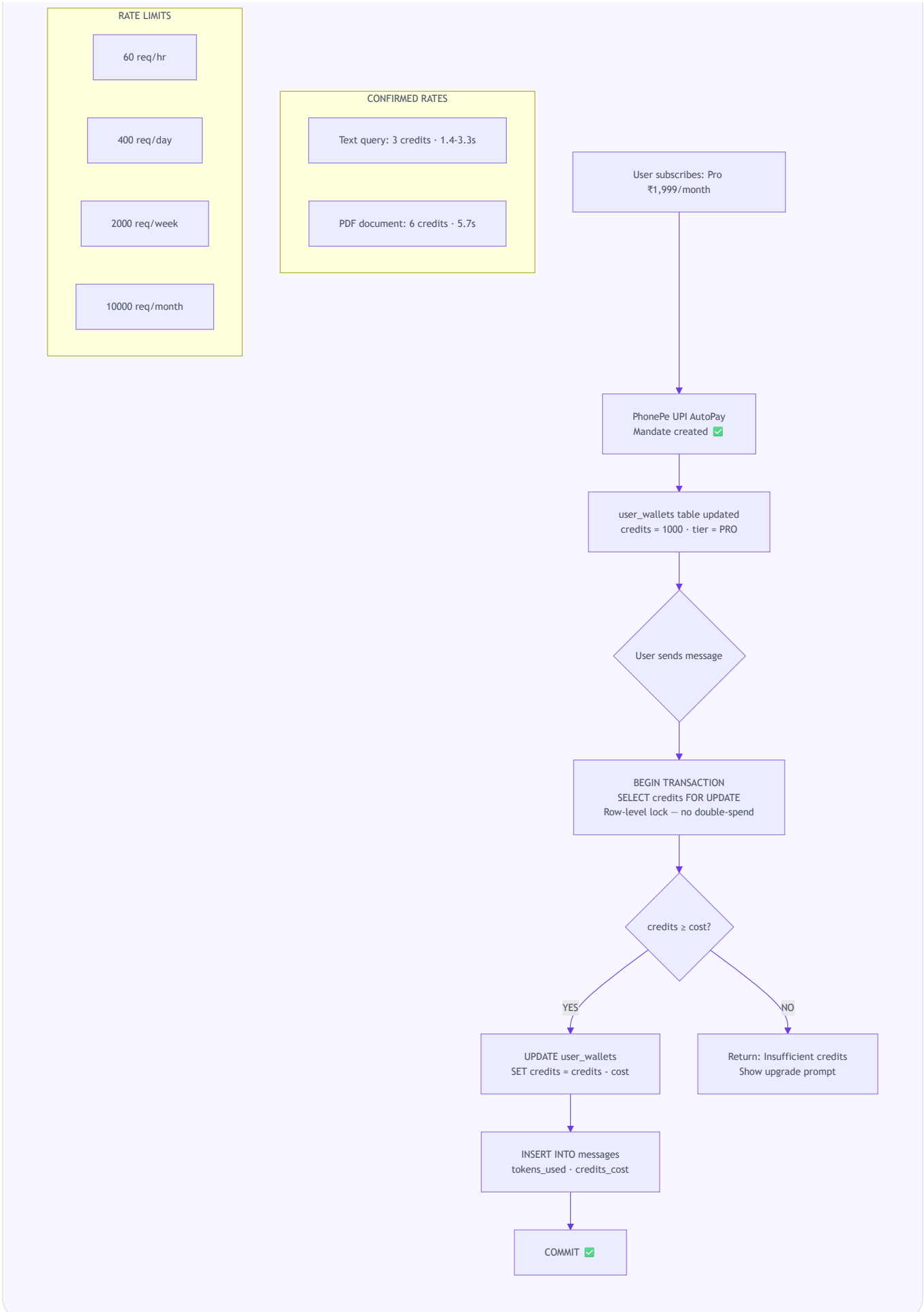


Anti-Hallucination Rulebook

Root Cause	Problem	Fix Required
Knowledge Gap	LLM doesn't know Grizon's own pricing	Populate + wire Pinecone company_docs RAG
Context Confusion	Multiple topics confuse the LLM	XML tags: `<DOCUMENT_CONTENT>`, `<RETRIEVED_KNOWLEDGE>`
OCR Noise	Hindi scans produce garbage text	5-step image preprocessing + Sarvam Vision
Prompt Ambiguity	Vague query → LLM guesses intent	Intent classification + entity extraction
High Temperature	Creativity causes invented facts	temp=0.1 for document Q&A mode
No Verification	Hallucinations reach user undetected	Wire ResponseVerifier LLM-as-Judge
Legal Citations	AI fabricates plausible case names	Indian Kanoon API mandatory for Legal Agent
IPC→BNS Confusion	Wrong section numbers post July 2024	Date-based law transition detection

11. Credit & Billing System Analysis

📐 FIGURE 11 — CREDIT TRANSACTION FLOW


RATE LIMITS
60 req/hr
400 req/day
2000 req/week
10000 req/month
CONFIRMED RATES
Text query: 3 credits · 1.4-3.3s
PDF document: 6 credits · 5.7s
User subscribes: Pro
₹1,999/month
PhonePe UPI AutoPay
Mandate created
user_wallets table updated
credits = 1000 · tier = PRO
User sends message
BEGIN TRANSACTION
SELECT credits FOR UPDATE
Row-level lock — no double-spend
credits ≥ cost?
YES
NO
UPDATE user_wallets
SET credits = credits - cost
Return: Insufficient credits
Show upgrade prompt
INSERT INTO messages
tokens_used · credits_cost
COMMIT

⚠️ **UI Confusion — Two Credit Pools:** "Plan credits: 6 of 1,000" vs "Credits remaining: 991" are shown without explanation. Users don't understand the difference. Add tooltip: *"Plan credits reset monthly. Wallet credits = your permanent spendable balance."*

12. Founder Recommendations

Top 5 Things to Do Right Now

1

FIX THE COMPOSER PUSH BUG — TODAY (2 HOURS)

Apply flex layout CSS fix. Pin composer to bottom. Messages area scrolls independently. After this fix, every single conversation becomes usable. Zero code risk — pure CSS.

2

POPULATE PINECONE + WIRE RAG — THIS WEEK (1 DAY)

Create pricing.txt, features.txt, faq.txt. Ingest to grizonai-knowledge (category: company_docs). Wire queryPinecone() to chatController.js. After this fix, the bot knows its own product.

3

WIRE RESPONSEVERIFIER + CONFIDENCE BADGES (2 DAYS)

Wire LLM-as-Judge to chat flow. Show [HIGH/MEDIUM/LOW CONFIDENCE] badge on every response. This delivers the core anti-hallucination promise that Grizon AI was built around.

4

LEGAL AGENT FOUNDATION — WEEK 3 (1 WEEK)

Indian Kanoon API + Legal RAG (BNS, BNSS, BSA, Constitution) + file_gen implementation + IPC→BNS date detection. Without these, the Legal Agent will hallucinate case citations.

5

SARVAM AI FULL INTEGRATION — MONTH 2 (2-4 WEEKS)

Saaras v3 STT + Bulbul v2 TTS + sarvam-translate + Sarvam Vision OCR. This single integration expands Grizon AI's addressable market from 250M English speakers to 1.4 Billion Indians.

Legal Agent vs. Competitors — Unique Value

Capability	Generic Legal AI	Grizon Legal Agent
Response format	Chat only	Chat summary + Full 10-section report document
Analysis depth	Statute lookup	6-stage structured reasoning workflow

Capability	Generic Legal AI	Grizon Legal Agent
Argument balance	One-sided information	Both sides argued objectively with precedents
India-specific	Generic international law	BNS/BNSS/BSA + 25 HC jurisdictions + limitation periods
Confidence rating	None	Strong / Moderate / Weak + High / Medium / Low
"Game-changer fact"	Not provided	Explicitly called out — the ONE fact that could change everything
Language support	English only	Hindi + 22 Indian languages via Sarvam (when wired)
Scanned documents	Usually fails	Sarvam Vision OCR — Hindi FIRs, court notices

Market Opportunity

Metric	Before Sarvam Integration	After Sarvam Integration
Addressable Indian Users	~250M (English only)	~1.4 Billion (all Indians)
Language Support	1 (English)	23 (22 Indian + English)
Voice-first Access	No	Yes (Saaras v3 STT)
Scanned Document Analysis	No	Yes (Sarvam Vision OCR)
Legal Reach	English-educated	All 60M+ pending court case parties

🚀 **Bottom Line:** Grizon AI has an excellent foundation. Payments work. Chat works. Document analysis works. The gaps — broken RAG, missing verification, no Hindi support — are all fixable within 4 weeks. The Legal Agent design is genuinely world-class for Indian legal AI; it just needs the right tools wired up. Fix the 3 critical bugs first, then launch the Legal Agent, then open up to 1.4B Indians with Sarvam.

Appendix — Technology Stack Reference

Technology	Role in Grizon AI	Reference
Sarvam Saaras v3	Hindi STT — voice input (22 Indian languages)	api.sarvam.ai
Sarvam Bulbul v2	Hindi TTS — voice output (abhilash/anushka voice)	api.sarvam.ai
sarvam-translate v1	22 Indian language translation (formal mode)	api.sarvam.ai
Sarvam Vision	Hindi/Devanagari OCR — scanned documents, FIRs	api.sarvam.ai
Indian Kanoon API	Free legal case search — SC + HC + Tribunal judgments	indiankanoon.org/api
Pinecone	Vector DB for RAG — company_docs + legal_docs indexes	pinecone.io
PyMuPDF (fitz)	Digital PDF text extraction — zero OCR errors	pymupdf.readthedocs.io
PaddleOCR	Mixed Hindi+English OCR — 88% accuracy	paddlepaddle.org
Redis + Sentence-Transformers	Semantic cache — cosine similarity ≥ 0.92	redis.io
PhonePe PG	UPI payments — SHA256 X-VERIFY checksum	developer.phonepe.com
Claude 3.5 Sonnet	Primary LLM (Project Brain co-primary)	anthropic.com
DeepSeek V3/R1	Fallback LLM (Project Brain co-primary)	api.deepseek.com
GPT-4o/5	Tertiary fallback LLM (Project Brain)	platform.openai.com
PostgreSQL	Primary DB — users, wallets, conversations, transactions	postgresql.org
Next.js + Tailwind	Frontend framework — React SSR	nextjs.org

Prepared by: Senior AI Systems Architect

Grizon AI

Master Technical Document — Confidential

Platform: Grizon v0.4.0 | August 7, 2026

Live session: app.grizonai.com (raj073032@gmail.com)